

Total No. of Questions : 4]

SEAT No. :

PD126

[6410]-448

[Total No. of Pages : 2

T.E. (Electronics Telecommunication) (Insem)
DIGITAL IMAGE PROCESSING
(2019 Pattern) (Semester-II) (Elective-II) (304195 A)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Explain functional blocks of Digital image processing in short. [6]

b) Consider two pixels a and b whose coordinates are (0,1) and (7,8). Calculate the Euclidian distance, chess board distance and city block distance. [3]

c) Explain the following connectivity between the pixels. [6]

4-connected

8-connected

M-connected

OR

Q2) a) What is Histogram of an image? Explain histogram of dark, bright and high contrast image. [5]

b) Compare RGB and YIQ color models. [5]

c) How much storage capacity is required to store an image with the size 1024×768 and 256 grey levels. [5]

P.T.O.

Q3) a) What is need of image enhancement? Explain categories of spatial domain image enhancement. [5]

b) Obtain histogram and histogram equalization of given image containing grey levels from 0 to 7. [5]

4	4	4	4	4
3	4	5	4	3
3	5	5	5	3
3	4	5	4	3
4	4	4	4	4

c) Compare point processing and mask processing. [5]

OR

Q4) a) Explain basic grey level transformation. [5]

b) Consider the following image, what will be the new value of a pixel (3,3) if smoothing is done using 3×3 mask. [5]

3	1	2	0	4
7	5	6	8	0
2	3	0	4	6
3	5	6	7	3
0	9	1	7	2

c) Explain Bit plane slicing and its importance. [5]

